PHYC10004

SCHOOL OF PHYSICS
SEMESTER 2 EXAMINATION

THURSDAY 9TH NOVEMBER 2017

EXAMINATION QUESTION PAPER

SEAT NUMBER										
STUDENT ID										

Do NOT write your name anywhere on your examination papers.

SUBJECT NAME:	PHYSICS 2: PHYSICAL SCIENCE AND TECHNOLOGY
SUBJECT CODE:	PHYC10004
EXAM DURATION:	3 hours NOT including reading time.
READING TIME:	15 MINUTES
TOTAL MARKS:	150 marks
NUMBER OF PAGES:	14 PAGES (INCLUDING THIS PAGE)

AUTHORISED MATERIALS

1. All students are authorised to have writing materials, and clear pencil cases in their possession.
2. Students are permitted to use only the University's authorised calculator
3. Students ARE NOT permitted to use: any dictionary.
4. Electronic devices – including lap tops, tablets, mobile phones and electronic dictionaries- are NOT authorised.

INSTRUCTIONS TO INVIGILATORS

1. This paper must REMAIN in the room and be collected at the end of the exam.
2. This Exam is NOT an open book exam. NO notes or books are permitted.
3. Students will require: this paper and TWO (2) script books.
4. Students are NOT permitted to annotate during reading time.
5. The final two pages of this exam paper contain formulae and data for the use of students. These pages may be detached, but students may NOT retain any part.

INSTRUCTIONS TO STUDENTS

1. This Exam is NOT an open book exam. NO notes or books are permitted.
2. Write your seat number and student ID number in the boxes at the top of this page at the START of WRITING TIME.
3. Attempt ALL questions. The total number of marks is 150.
4. Write your answers: on the RULED PAGES of your script book/s ONLY. Blank pages are for rough working out and will not be marked.
5. Two pages of formulae and data are attached to the paper. Detach these pages for use. Hand the pages to a supervisor with your completed examination paper at the end of writing time.

6. ALL communications-enabled & network accessible devices MUST be switched OFF & placed under your seat.

EXAMINATION REPRINTS

1. This paper may be lodged with the library.

Student: you are reminded that:

- Failure to follow examination instructions constitutes misconduct.
- You MUST NOT have any unauthorised materials or items with you that could be used to cheat, be unscholarly and/or gain an unfair advantage over other students.
- Unauthorised item/s must be declared and, wherever possible, surrendered to a supervisor.
- Penalties apply for misconduct.
- Any student found cheating, suspected of cheating, who attempts to cheat and/or who enables others to cheat in an exam is engaging in academic misconduct.
- Students found to have engaged in general or academic misconduct face penalties ranging from exclusion from the University to a fine, losing credit for the examination and/or the termination of their enrolment.
- A supervisor will submit a report in writing about incidents of alleged misconduct to the University.
- If you do not have photo ID with you, you will need to report to the Chief Supervisor at the end of the exam and have your photo taken.

Before your exam

- Please place your student card on top left hand corner of your desk.
- Empty your pockets and place all excess items under your desk or in the bin.
- Please read and review your question paper during reading time. Unless you are permitted to annotate your paper, you must not write during reading time.
- If you need assistance from a supervisor, including materials or a toilet break, raise your hand. Wait until the supervisor attends to you. Do NOT leave your seat without permission.
- Wireless communications, mobile phones and any device other than an authorised device must be switched off and placed under your desk.
- If you do not have your student card, please place another form of photo identification such as your passport or driver's licence, in the top left hand corner of the desk, instead.

During your exam

- You may not leave in the first half hour or the last fifteen minutes of any University of Melbourne examination.
- You must not communicate with, receive assistance from, or copy from the paper of another student, at any time during your exam.
- You must obey the reasonable instructions of the examination supervisors.
- Supervisors will inspect desks and materials that you access. Unauthorised material will be confiscated. You may be reported for possessing unauthorised items.

In an emergency

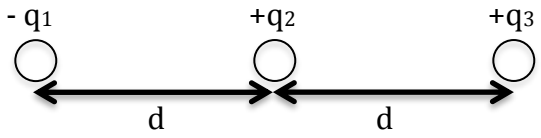
- In the event of fire, or an emergency evacuation, leave all examination material where it is. Make your way to the nearest exit. Assemble away from the building, in line of sight of a fire warden who will be wearing a hard hat. Await further instructions.

Signals and warnings at the end of your exam

- A warning signal will be given ten minutes ahead the end of your examination. After the signal, you are NOT permitted to leave the venue and must wait until dismissed.
- At examination's end a final signal will be given. The Chief Supervisor will order 'pens down'. You must cease writing and put your pen down, when ordered to do so.

Question 1 Electrostatics **(Total 8 marks)**

Three charges, $-q_1$, $+q_2$ and $+q_3$ are arranged in a line as shown. The charges are a distance d apart. In this arrangement q_3 is in equilibrium while q_1 and q_2 are fixed.



a) Draw and label all of the forces acting on q_3, including relative direction and magnitude.

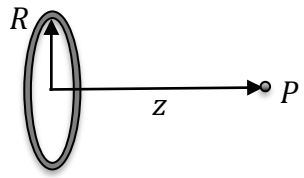
b) In this arrangement show that $q_1 = 4\ q_2$.

c) The charge q_2 is now removed, q_1 and q_3 have a charge of -10 nC and +2 nC respectively and the distance d is 5 cm, what is the electric potential at the position mid-way between the two charges?

(2 + 3 + 3 = 8 marks)

Question 2 Electrostatics **(Total 6 marks)**

An insulator in the shape of a thin ring, radius R, is given a total charge, Q, equally distributed over the ring.



a) At a point P, a distance of z from the centre of the ring and lying on the axis of the ring as shown above, show that the electric potential due to the ring of charge Q is

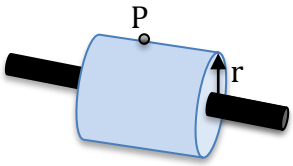
$$V = \frac{1}{4\pi\varepsilon_0}\frac{Q}{\sqrt{z^2 + R^2}}$$

b) Using the electric potential, write an expression for the magnitude of the electric field at P.

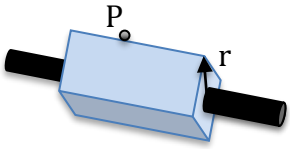
(3 + 3 = 6 marks)

Question 3 Guass's Law **(Total 14 marks)**

Jean and Yang wish to determine the electric field of a very long charged wire, as shown below. The wire has a linear charge density of λ C m^{-1}. Jean suggests that a cylindrical Gaussian surface would be the most useful way to determine the electric field of the wire at point P. Yang disagrees and suggests a rectangular prism.



Jean's Gaussian surface



Yang's Gaussian surface

a) Explain, in terms of Gauss's Law, why a cylindrical Gaussian surface is the most appropriate geometry for this situation and not a rectangular one.

b) Using Gauss' Law, find an expression for the electric field of the wire at a distance *r*.

c) If the charge on the wire is positive and the linear charge density is 20 nC m^{-1}, what is the magnitude of the electric field of the wire at *P* where *P* is a distance of 8.0 cm from the wire?

d) Draw a diagram of the electric field of the wire from the following viewpoints. Indicate whether the electric field is uniform or non-uniform.

Side View

End View

(4 + 5 + 2 + 3 = 14 marks)

Question 4 Capacitors **(Total 13 marks)**

BestMed, a medical technology company, is designing a new portable defibrillator for reviving patients whose heart has stopped. The capacitor in the device will be charged to 2000 V and will be required to deliver 400 J of energy on discharge.

a) Calculate the capacitance required for the capacitor in the device.

To fit inside the casing, the capacitor plates must have an area of 5 m^2 and separated by an insulator with thickness of 0.05 mm.

b) Show that the dielectric constant must be approximately 225 for the insulator between the capacitor plates to deliver the intended energy to the patient.

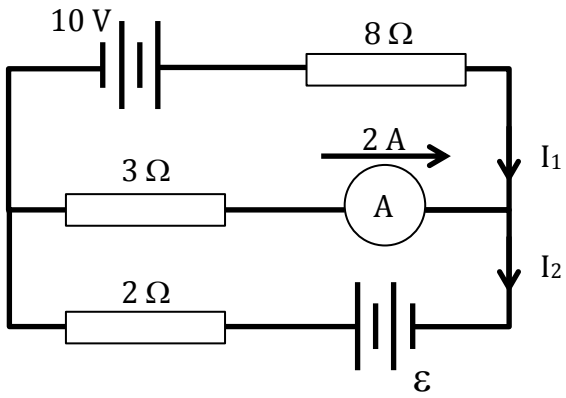
A dielectric material with the desired dielectric constant has a dielectric strength of 4×10^6 V m^{-1}.

c) Will this material be a suitable candidate for the machine? Why, why not? Support your statement with appropriate calculations.

(3 + 5 + 5 = 13 marks)

Question 5 Circuits **(Total 8 marks)**

A circuit is set up with three resistors and two ideal batteries as in the diagram below. One of the batteries has an emf 10 V, but the emf of the second battery is unknown. An ammeter shows that 2 A is flowing through the 3 Ω resistor from left to right.



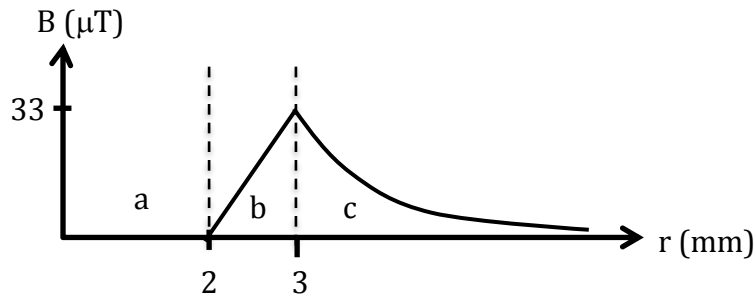
a) Using Kirchoff's junction and loop laws, determine the emf, ε, of the second battery.

b) In the circuit above the 3 Ω resistor is replaced with a globe while a current of 2 A is still measured by the ammeter. The voltage across the globe is measured to still be 6 V and therefore the resistance of the globe is 3 Ω according to the equation $V = iR$. Can we assume the globe is ohmic? Explain your answer.

(5 + 3 = 8 marks)

Question 6 Magnetic Field **(Total 13 marks)**

A certain shaped wire conductor carrying a current *I*, produces a magnetic field *B* versus radial distance from the centre of the wire, *r* as shown in the graph below.



a) Draw a diagram and describe the shape of the wire that produces a magnetic field distribution as shown in the graph above. Indicate section a, b and c of your wire and any significant distances.

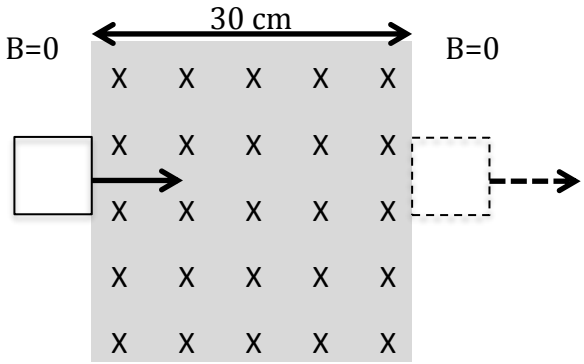
b) If the wire's *overall* shape is cylindrical, use Ampere's law and the graph above to determine the size of the current through the wire.

c) When Maxwell derived his four equations for Electromagnetism, he added an extra term to Ampere's Law. Explain the significance of this extra term and give an example of where we can find evidence for its existence.

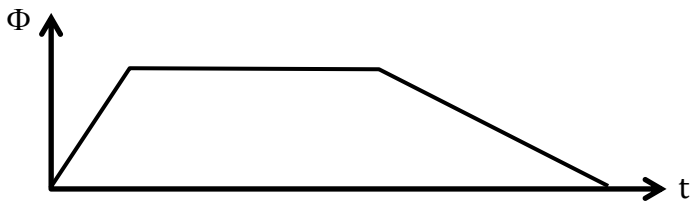
(5 + 5 + 3 = 13 marks)

Question 7 Faraday's Law **(Total 13 marks)**

A square loop of copper wire moves through a magnetic field, B, as shown in the diagram below. The loop is perpendicular to the direction of the magnetic field, which is into the page.



As the loop moves into, through and out of the magnetic field the magnetic flux versus time is shown in the graph.



a) Sketch the resulting induced emf graph for the loop. Label any areas or points of interest.

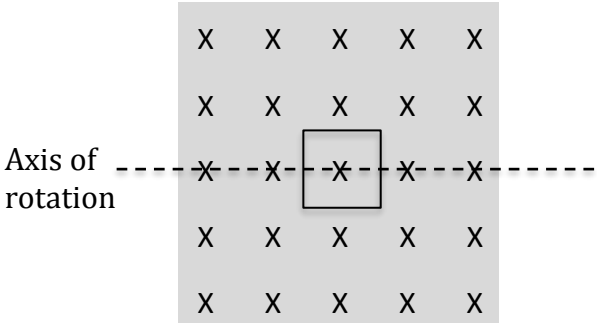
b) From the graph above, which of the following statements (A-E) about the motion of the loop is correct?
   A) The loop travels into, through and out of the field at a constant rate.
   B) The loop travels into the field at a faster rate than through and out of the field.
   C) The loop travels into the field at a faster rate than out of the field, but we do not have enough information to determine its speed through the field.
   D) The loop travels into the field at a slower rate than through and out of the field.
   E) The loop travels into the field at a slower rate than out of the field, but we do not have enough information to determine its speed through the field.

The loop is now placed in the centre of the magnetic field which now changes with time according to $B = 0.7-0.4t^2$. The loop has a side length of 10 cm.

c) What is the magnitude of the induced emf in the loop after 2 seconds?

Question 7 (continued)

d) The magnetic field is now a constant B = 0.7 T and the loop is attached to a turbine so it can spin around the axis of rotation as shown below. What must the frequency of rotation be to produce a maximum emf of 3 V?



e) Provide ONE way the emf induced in the loop can be increased by only changing the properties of the loop?

(3 + 1 + 5 + 3 + 1 = 13 marks)

Question 8 Fluids **(Total 14 marks)**

A 70 kg ancient Roman statue lies at the bottom of the Mediterranean Sea. Its volume is 3.0×10^4 cm^3.

a) How much force is needed to lift it slowly up to the surface?

b) What is the force needed to hold it stationary once the statue is outside the water?

c) The statue holds a shield. Archaeologists want to know if the shield is of gold or gold painted lead. Knowing that the shield has a mass of 14.70 kg on the boat and 13.74 kg when submerged, what is the shield made of?

d) Water is flowing in a fire hose with a speed of 1.0 m s^{-1} and a pressure of 200 kPa. At the nozzle the pressure decreases to atmospheric pressure (101.3 kPa), there is no change in height. Use the Bernoulli equation to calculate the speed of the water leaving the nozzle.

(Densities: sea-water 1.03×10^3 kg m^{-3}, gold 19.3×10^3 kg m^{-3}, lead 11.3×10^3 kg m^{-3})

(4 + 2+ 4 + 4 = 14 marks)

Question 9 Heat and Work **(Total 15 marks)**

a) If 400g of ice at -2 ^{0}C is placed in 1 kg of water at 21 ^{0}C what is the end product when equilibrium is reached?

b) It takes 487.5 J to heat 25 grams of copper from 25 °C to 75 °C. What is the specific heat in Joules/g·°C?

c) Calculate the internal energy change of 1 mol of an ideal monoatomic gas for the following two processes..

i) 1500 J of heat is added to the gas. The gas does no work (and no work is done on the gas).

ii) 1500 J of work is done on the gas (and the gas does no work). No heat is added or taken away.

d)

i) Calculate the volume of 62.5 mol of oxygen at a pressure of 1.00×10^5 Pa at T = 27 ^{0}C?

ii) If the temperature of the gas in part c) is reduced to -10 ^{0}C but its volume is held constant, what is the new pressure? (Treat oxygen as an ideal gas).

(4 + 3 + (2 + 2) + (2 + 2) = 15 marks)

Question 10 Particle wave duality **(Total 13 marks)**

c) The work function for sodium and caesium are 2.28 eV and 2.14 eV, respectively.

i) Find the threshold frequency sodium and caesium.

ii) For incident photons of a given frequency, which metal will give a higher maximum kinetic energy for the electrons? Explain your reasoning

b) A Ferrari with a mass of 1200 kg approaches a freeway underpass that is 10 m across. At what speed must the car be moving to have a wavelength that may be diffracted by the underpass (single slit)? State your assumptions.

c) We wish to measure the wavelength and position of a photon simultaneously. The wavelength of the photon is 600 nm and is known with an accuracy $\Delta p/p = 10^{-6}$. What is the minimum uncertainty in the position of the photon?

(3 + 3 + 4 + 3 = 13 marks)

Question 11 The black body **(Total 8 marks)**

a) The Sun can be assumed to be a black body.

i) Estimate the temperature of the surface of the Sun, knowing that the Sun emits light in the visible spectrum, with $\lambda_{max} = 500$ nm.

ii) Assuming the Sun is a sphere, of 6.09×10^{18} m^2 surface area, how much power does it radiate?

b) Although the Sun emits a spectrum of waves all having different wavelengths, model its whole power output as carried by photons of wavelength λ_{max}:

i) Find the energy of one photon.

ii) Find the number of photons it emits each second.

(2 + 2 + 2 + 2 = 8 marks)

Question 12 The atom **(Total 13 marks)**

a) The Balmer series for the hydrogen atom corresponds to electronic transitions that terminate in the state of quantum number $n = 2$, For the following parts of this question, use Bohr's atomic theory.

   i) Find the energy of the longest wavelength photon emitted.
   ii) Calculate the radius of the Hydrogen atom for $n = 10$?
   iii) Calculate the speed of an electron in in an $n = 10$ state in a hydrogen atom?

b) On the basis of Bohr's model of the atom, explain why all emission lines are not observed in absorption.

c) Assume that the chromium atom ($Z = 24$) can be treated as a hydrogen-like atom with *a small modification of the Z-value*

   i) Calculate the energy difference transition of an electron in a chromium atom from the $n = 2$ state to the $n = 1$ state. For this calculation take $Z = 23$ (to allow for the shielding effect of a single 1s electron).

   ii) In the *Auger process*, this energy is not emitted as a photon. Instead, it is transferred to an electron in the $n = 4$ state that is ejected by the atom. Use the Bohr theory to find the kinetic energy of the emitted Auger electron. For the ionisation energy of the $n = 4$ state, take $Z = 5$ (to allow for the shielding of the inner electrons).

(2 + 2 + 2 + 2 + 2 + 3 = 13 marks)

Question 13 The nucleus, nuclear decays **(Total 12 marks)**

A laboratory has 1.49 μg of pure ${}^{13}_{7}N$. It has a half-life of 600 s.

a) Show that the number of atoms initially present is equal to 6.90×10^{16}.

b) Find the initial activity for the ${}^{13}_{7}N$.

c) How many atoms remain after 1 hour?

d) Find the radius of a ${}^{13}_{7}N$ nucleus.

e) Write the decay equation that produces a ${}^{13}_{7}N$ atom by β emission?

f) Consider two heavy nuclei, X and Y, with similar mass numbers. If X has the higher binding energy, which nucleus tends to be more unstable? Briefly explain your reasoning.

(2 + 2 + 2 + 2 + 2 + 2 = 12 marks)

------END OF EXAM------

Formula Sheet and Useful Data for PHYC10004

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Constants

Mass of electron, m_e	9.11×10^{-31} kg	Bohr radius, a_B	0.529×10^{-10} m
Mass of proton, m_p	1.67×10^{-27} kg	Rydberg constant, R	1.097373×10^{7} m^{-1}
Electron charge magnitude, e	1.6×10^{-19} C	Stephan-Boltzmann constant, σ	5.67×10^{-8} W m^{-2} K^{-4}
Permeability of free space, μ_o	$4\pi \times 10^{-7}$ T m A^{-1}	Boltzmann constant, k	1.38×10^{-23} J K^{-1}
Permittivity of free space, ε_o	8.85×10^{-12} C^2 N^{-1} m^{-2}	Speed of light in vacuum, c	3.00×10^{8} m s^{-1}
Universal gas constant, R	8.31 J K^{-1} mol^{-1}	Gravitational acceleration, g	9.8 m s^{-2}
Avagadro's number, N_A	6.02×10^{23} mol^{-1}	1 eV = 1.6×10^{-19} J	
Specific heat capacity of ice	2220 J K^{-1} kg^{-1}	Planck's constant, $h = 6.63 \times 10^{-34}$ J s = 4.14×10^{-15} eV s	
Specific heat capacity of water	4187 J K^{-1} kg^{-1}	Latent heat of fusion for water	333 kJ K^{-1}

Formulae

Electromagnetism		
$\boldsymbol{F} = \frac{q_1 q_2}{4\pi\varepsilon_o r^2}\hat{r}$	$\boldsymbol{E} = \frac{\boldsymbol{F}}{q}$	$V(P) = \int_P^{\infty} \boldsymbol{E}.d\boldsymbol{s} = \frac{U(P)}{q}$
$W = -\Delta U$	$\boldsymbol{E} = \frac{q}{4\pi\varepsilon_o r^2}\hat{\boldsymbol{r}}$	$V = \frac{q}{4\pi\varepsilon_o r}$
$\oint \boldsymbol{E}.d\boldsymbol{A} = \frac{q}{\varepsilon_o}$	$\Delta V = -\int_{P_2}^{P_1} \boldsymbol{E}.d\boldsymbol{s}$	$\boldsymbol{p} = q\boldsymbol{d}$ $\boldsymbol{\tau} = \boldsymbol{p} \times \boldsymbol{E}$ $U = -\boldsymbol{p}.\boldsymbol{E}$
$V = iR$	$P = iV = \frac{V^2}{R} = i^2 R$	$\frac{V_S}{V_p} = \frac{N_S}{N_p}$
$C = \frac{Q}{V}$	$U = \frac{1}{2}qV = \frac{1}{2}CV^2 = \frac{1}{2}\frac{q^2}{C}$	$C = \frac{A\varepsilon_o}{d}$
$\varepsilon = \kappa\varepsilon_0$	$U_E = \frac{1}{2}\kappa\varepsilon_o E^2$	$i = \frac{dq}{dt} = enAv_d$
$\boldsymbol{F} = q\boldsymbol{v} \times \boldsymbol{B}$	$\boldsymbol{F} = i\boldsymbol{l} \times \boldsymbol{B}$	$\oint \boldsymbol{B}.d\boldsymbol{A} = 0$
$B = \mu_o n i$	$\Phi_E = \int \boldsymbol{E}.d\boldsymbol{A}$	$\Phi_B = \int \boldsymbol{B}.d\boldsymbol{A}$
$\xi = -\frac{d\Phi_B}{dt}$	$\oint \boldsymbol{E}.d\boldsymbol{s} = -\frac{d\Phi_B}{dt}$	$\oint \boldsymbol{B}.d\boldsymbol{s} = \mu_o i + \mu_o i_D$
$i_D = \varepsilon_o \frac{d\Phi_E}{dt}$	$\mu = i \times (area\ of\ loop)$	$\boldsymbol{\tau} = \boldsymbol{\mu} \times \boldsymbol{B}$ $U = -\boldsymbol{\mu}.\boldsymbol{B}$

Formula Sheet and Useful Data for PHYC10004

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Fluids and Thermal Energy		
$R = A_1 v_1 = A_2 v_2$	$p + \rho g h + \frac{1}{2}\rho v^2 = const$ $p = p_A + \rho g h$	$\frac{F}{A} = Y\frac{\Delta L}{L}$
$pV = nRT$	$\rho = \frac{M}{V}$	$Q = mL$ $Q = mc\Delta T$
$pV^\gamma = const$ $\gamma = \frac{c_p}{c_V}$	$P = \sigma\varepsilon A T^4$	$\Delta U = Q + W$
$W = -\int p dV = -p\Delta V$	$W = -nRT\ln\frac{V_f}{V_i}$	$v_{RMS} = \sqrt{\frac{3kT}{m}}$
Modern Physics		
$\lambda_{MAX} T = 2898\ \mu m\ K$	$hf = K_{MAX} + \phi$	$E = hf = \frac{hc}{\lambda}$
$\lambda = \frac{h}{p}$	$c = f\lambda$	$\Delta x \Delta p \geq \frac{\hbar}{2}$
$\Delta\lambda = \frac{h}{mc}(1 - \cos\theta)$	$N(t) = N(0)e^{-\lambda t}$	$\lambda = \frac{\ln 2}{\tau_{1/2}}$
$E_n = -\frac{h^2}{8mL^2}n^2$	$r_n = \frac{4\pi\epsilon_o \hbar^2}{me^2}n^2 = a_o n^2$	$v_n = c\alpha_{em}\frac{1}{n}$
$E_n = -\frac{13.6\ Z^2}{n^2}$	$a_o = \frac{\epsilon_o h^2}{\pi m e^2} = 52.92\ pm$	$\alpha_{em} = \frac{e^2}{4\pi\epsilon_o \hbar c} \approx \frac{1}{137}$
$f = cR\left(\frac{1}{n_2^2} - \frac{1}{n_1^2}\right)$	$r = (1.2 \times 10^{-15} m)A^{\frac{1}{3}}$	$\frac{B.E.}{c^2} = (A - Z)m_n + Zm_p - m_{nucleus}$